

Reproduction:

Reproduction is the process by which living organisms produce more living organisms of its own kind.



Mode of Reproduction:

There are two main types of reproduction in living organisms.

1. Asexual Reproduction: The process of reproduction in which new individuals are produced by single parent. E.g. amoeba, hydra etc

2. Sexual Reproduction: The process of reproduction in which two individuals are involved to produce a new individual. E.g. Human, tiger etc

Sexual Reproduction:

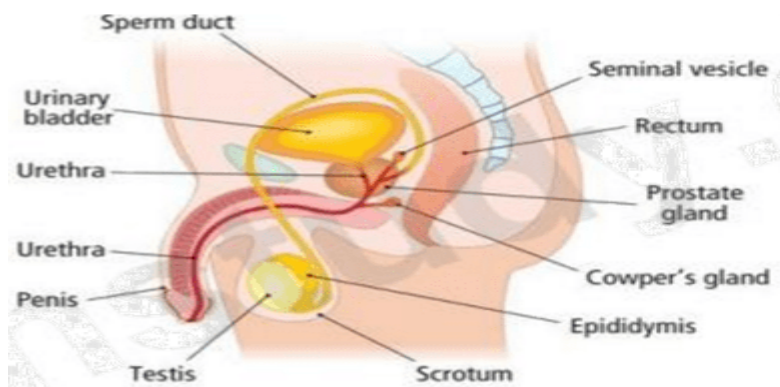
It may be defined as the production of offspring's (new individuals) by the fusion of male and female gamete to form a zygote, which develops into a mature organism.

Reproductive Organs in Humans:

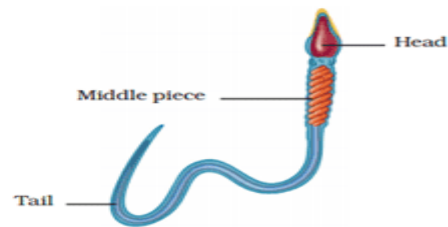
1. Male Reproductive Organs:

A pair of testes (singular, testis), two sperm ducts and a penis, these are the male reproductive organs. The testes produce the male gametes called sperms. Though sperms are single celled, very small in size, each has a head, a middle piece and a tail.





Male reproductive System



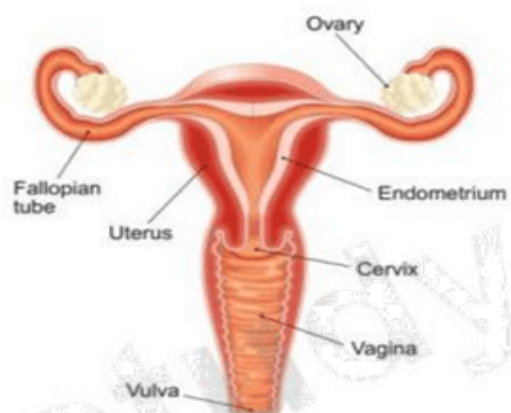
Structure of Sperm



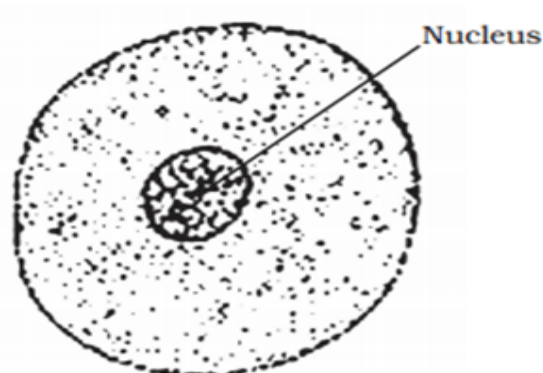
Sperm

Female Reproductive Organs:

A pair of ovaries, oviducts (Fallopian tubes) and the uterus, these are the female reproductive organs. Ovary produces female gametes called ova (eggs). Oviduct transports eggs from ovaries to the uterus. Uterus is the part where development of the baby takes place. An egg is also a single cell.



Female reproductive organs in humans

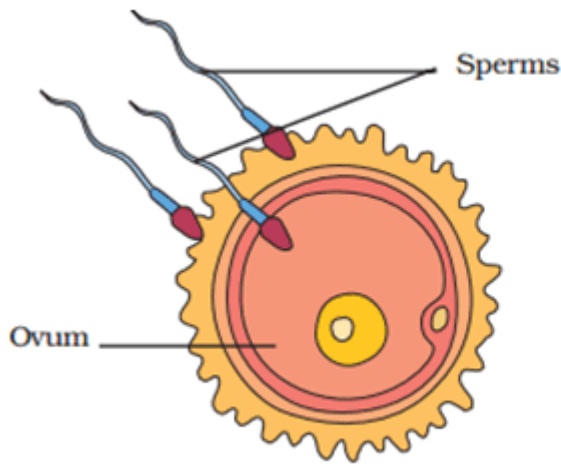


Human Ovum

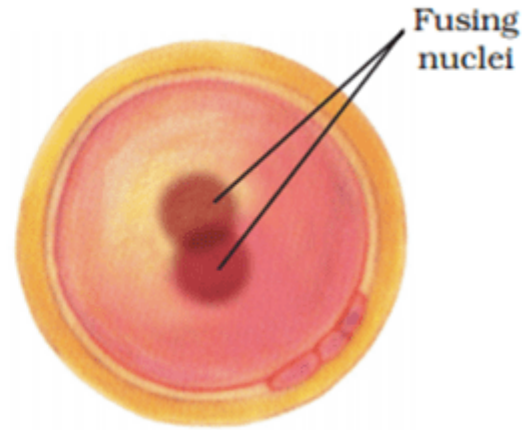


Fertilization:

The fusion of a male gamete (Sperm) with a female gamete (Ovum) is called fertilization. Zygote is formed after fertilization.



Fertilization

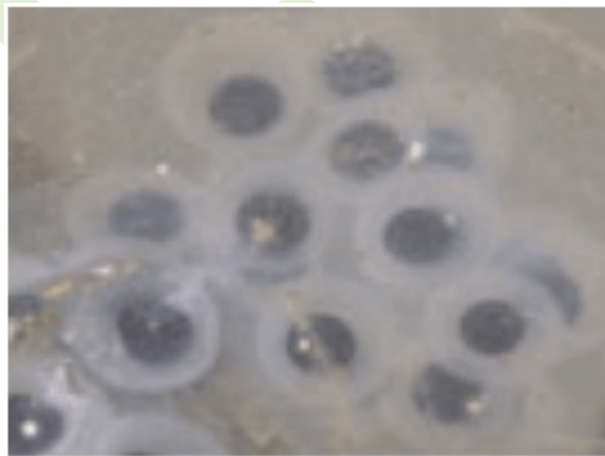


Zygote

Types of Fertilization: There are two types of fertilization in animals, external fertilization and internal fertilization.

1. Internal Fertilization: When fertilization takes place inside the animal's body, it is called internal fertilization. Internal fertilization occurs in many animals including humans, cows, dogs and hens.

2. External Fertilization: When fertilization (fusion of a male and a female gamete) takes place outside animal's body, it is called external fertilization. It is very common in aquatic animals such as fish, starfish, frog etc.

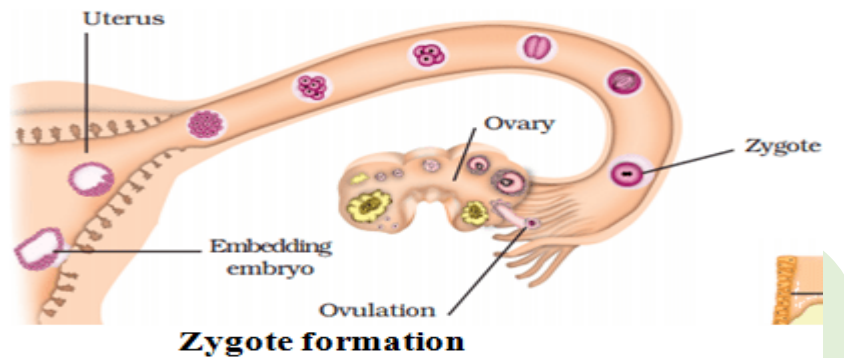


Eggs of frog



Development of Embryo: Development of embryo takes place in the female reproductive system through following steps:

1. Every month, a single egg comes out of the ovary and reaches the fallopian tube of female reproductive system.
2. During copulation, sperms reach the Fallopian tube where a sperm fertilizes the egg. This results in formation of zygote.

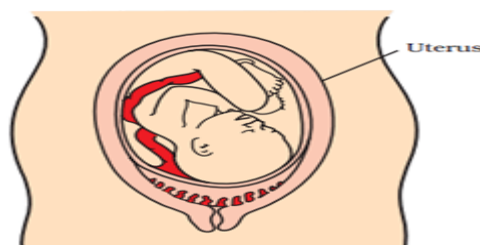


3. The zygote divides repeatedly to give rise to a ball of cells. The cells then begin to form groups that develop into different tissues and organs of the body. This developing structure is termed an embryo.



Ball of cells (enlarged)

4. The embryo gets implanted in the wall of the uterus and further develops the body parts such as hands, legs, head, eyes, ears, etc.
5. The stage of the embryo in which all the body parts can be identified is called a foetus. When the development of the foetus is complete, the mother gives birth to the baby.



Foetus in the uterus



Formation of Egg Shell in Hens:

Internal fertilization takes place in hens. After fertilization, the zygote divides continuously and moves to the oviduct. As it travels down, many protective layers are formed around it. The hard shell in a hen's egg is one such protective layer. After the hard shell is formed, the hen finally lays the egg. The embryo takes about 3 weeks to develop into a chick. The hen sits on the eggs to provide sufficient warmth. After the chick is completely developed it bursts open the egg shell.

Viviparous and Oviparous Animals:

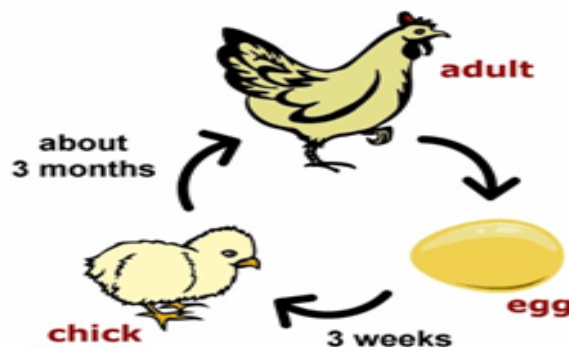
1. The animals which give birth to young ones are called viviparous animals. Examples- Human being, dog, cows.
2. Those animals which lay eggs are called oviparous animals. Examples – hen, frog etc.

IVF (In Vitro Fertilization):

The fertilization carried out in laboratory is called in-vitro fertilization. This technique is done in the women's, who can't bear babies. The sperm and egg are placed in the test tube until zygote is formed. Once the zygote is formed it is placed in the mother's uterus for further development into full baby. Babies born through this technique are called test-tube babies.

Young Ones to Adults:

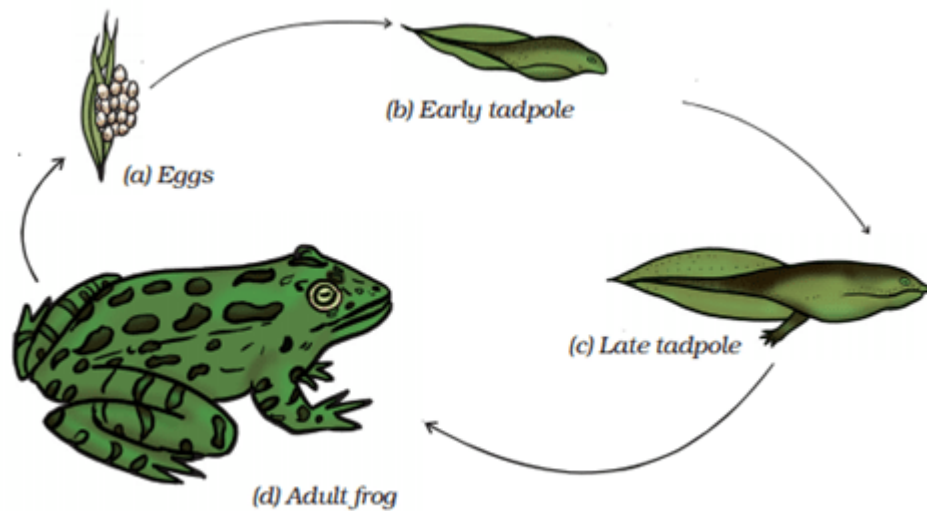
1. Direct Development: When the young ones of an animal resemble the adult, then direct development takes place, e.g. hen, man, monkey, etc.



Life Cycle of Hen

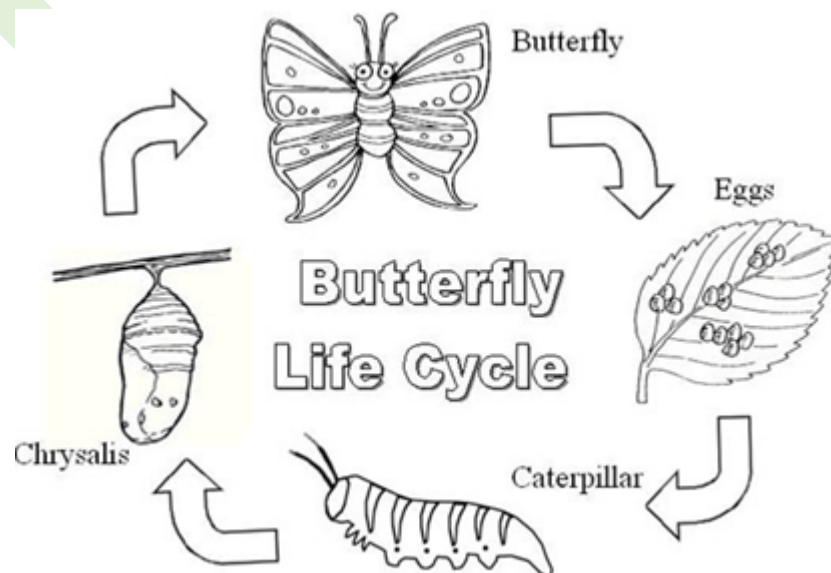


2. Indirect Development: When the young ones of an animal do not resemble the adult, then indirect development takes place, e.g. frog, butterfly, silk moth, etc.



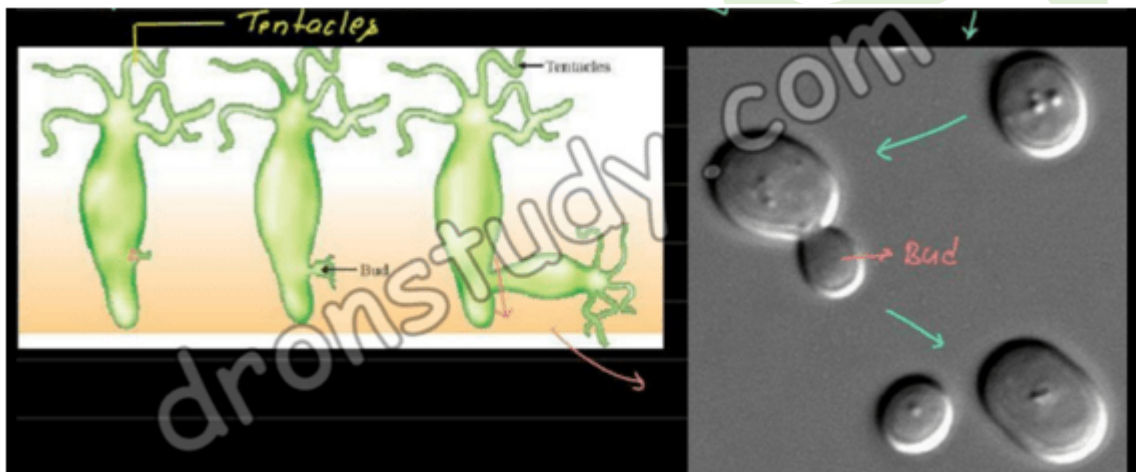
Life cycle of frog

3. Metamorphosis: In case of indirect development, transformation of young ones into adult through drastic changes is called metamorphosis. Larva of butterfly undergoes metamorphosis to become a butterfly. A tadpole undergoes metamorphosis to become a frog.



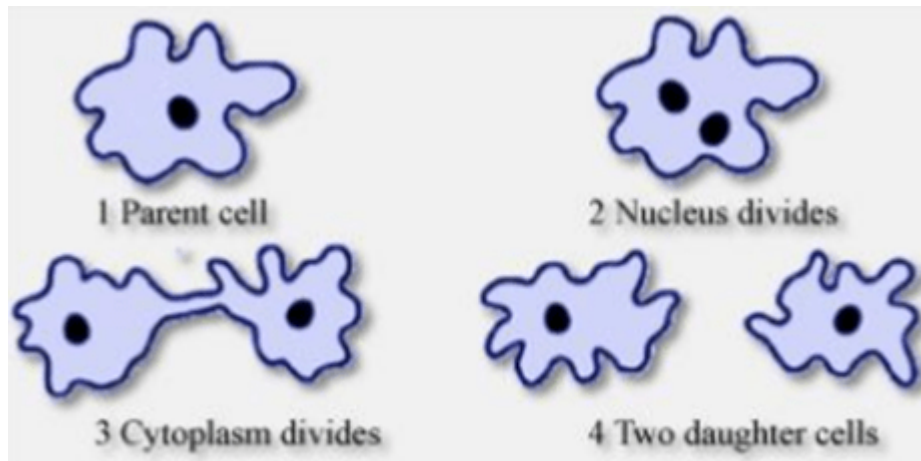
Asexual Reproduction Methods: The process of reproduction in which new individuals are produced by single parent. Example – amoeba, bacteria, hydra etc

1. Budding: In this type of asexual reproduction, a small bud or bulge develops on the body. After developing the bud, it gets detached from the parent's body to begin life as a new individual. Since new individuals develop from the buds, this type of asexual reproduction is called budding Examples: Hydra and sponges.



2. Binary Fission: In this method, an organism divides and forms two daughter cells. First the nucleus divides and forms two daughter nuclei. Then the cytoplasm in the mother cell divides into two daughter cells. This leads to the formation of the two daughter cells each having a nucleus and its own cell organelles which then develop into a fully formed adult. Example- paramecium, amoeba etc.





Binary fission in Amoeba

Cloning:

1. Cloning is process of production of an exact copy of a cell, any other living part, or a complete organism. Cloning process of an animal was successfully performed for the first time by Ian Wilmut and his colleagues at the Roslin Institute in Edinburgh, Scotland. They cloned successfully a sheep named Dolly.



(a) Finn Dorsett



(b) Scottish blackface ewe



(c) Dolly

